



Lehrstuhl für Multimediatechnik
und Signalverarbeitung
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Master Thesis

for

Mr. Jiahui Shen

Investigating the Impact of Point Cloud Upsampling on 3D Object Detection Performance

Untersuchung des Einflusses von Punktwolken-Upsampling auf die Leistungsfähigkeit der 3D-Objekterkennung

A big topic of research in the field of point cloud enhancement is the upsampling of point clouds. The goal of the upsampling methods is to create a more dense and uniform point cloud. The idea is to do upsampling as a first step before other tasks like shape classification and object detection are performed. Despite the upsampling task being seen as a preprocessing step, most works only evaluate their method by comparing the resulting point cloud with the ground truth using metrics like Chamfer Distance and point-to-surface distance.

Therefore, this thesis aims to investigate, which improvements can be achieved in object detection tasks when upsampling is done in a preprocessing step. For this analysis, multiple pipelines of upsampling methods and object detection networks will be build and compared with each other. Hereby, deep learning based and optimization based upsampling are assessed comparatively.

The research examines the extent to which upsampling methods can enhance object detection. Additionally, it investigates, whether the classic evaluation metrics on point cloud upsampling are good indicators for the improvements the methods can achieve in the object detection performance. We also want to explore, which type of upsampling method provides the most benefit.

As part of this work, a thorough literature review is required to examine the state-of-the-art upsampling and object detection techniques. In addition, the thesis is expected to provide clear and comprehensive documentation of the research, methodology, and findings.

Start: 01.03.2026

End: 01.09.2026

A handwritten signature in blue ink, reading 'A. Kaup', is written over a horizontal line. Below the line, the text 'Prof. Dr.-Ing. A. Kaup' is printed in a black, sans-serif font.

Prof. Dr.-Ing. A. Kaup